Pituitary Disorders III: Posterior Pituitary: Diabetes Insipidus, SIADH & Hyponatraemia

Internal Medicine Department | Source: `11- posterior pituitary disorders` (27 slides)

> **Legend:** ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.
>
> 🔗 **Bridge from earlier lectures:**
>
> - **Pituitary I–II:** suprasellar tumour extension, surgery and hypopituitarism causes all damage the **hypothalamus/stalk → central DI**.
> - **Hyperparathyroidism: hypercalcaemia → "calcium diabetes" (nephrogenic DI)**. This lecture's case.
> - **Lung cancer: small (oat) cell carcinoma → SIADH**.
> - **Physiology (membrane transport, body fluids):** osmolality and water channels (**aquaporin-2**) explain the whole topic.
> - **Hypertension:** thiazides are reused here to treat nephrogenic DI.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Pathophysiology and etiology of posterior pituitary disorders	ADH physiology; causes of central vs nephrogenic DI; causes of SIADH
Knowledge	Manifestations, complications, management	Polyuria; hyponatraemia symptoms; DDAVP vs thiazide
Skill	⭐ Interpret the **water deprivation test**	Central vs nephrogenic vs psychogenic
Skill	⭐ Diagnostic approach to **hyponatraemia**	Osmolality → volume status → urine Na
Skill	Emergency plans	Severe hyponatraemia → hypertonic saline, slow correction

Lecture flow: Posterior pituitary physiology → Control of ADH → **Diabetes insipidus** (definition, causes, clinical picture, investigations, water deprivation test, treatment) → DDx of polyuria → **SIADH** (causes, clinical picture, labs, criteria, treatment) → **Hyponatraemia algorithm** → Case.

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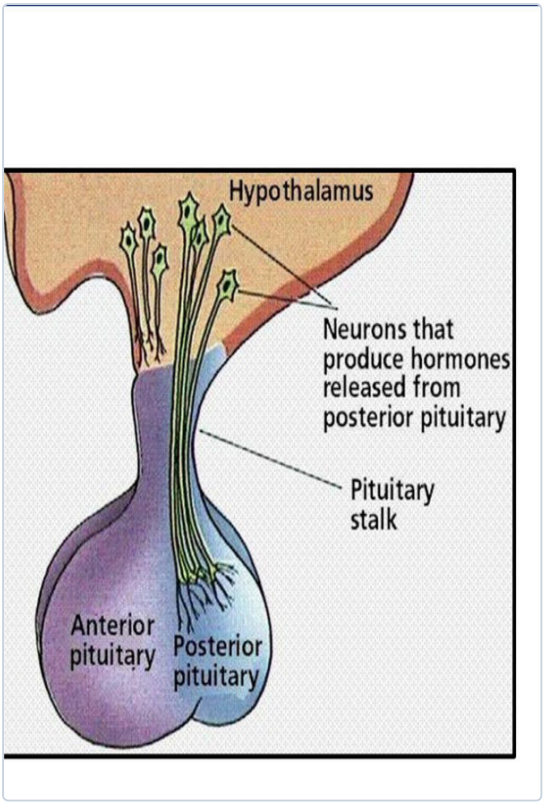
2. Main Content (Lecture Order)

2.1 Posterior Pituitary Physiology (Slides 4–5) ⭐

- **ADH (vasopressin, AVP)** is made in the **supraoptic** nucleus; **oxytocin** in the **paraventricular** nucleus.
- Both travel **down the axons** to the posterior pituitary for storage and release.

Hormone	Action
Oxytocin	Contracts **myoepithelial cells** of the breast → **milk ejection (let-down)** (*[added]* and uterine contraction)

Hormone	Action
⭐ **ADH / AVP**	↑ **water reabsorption in the collecting tubules** (*[added]* V2 receptor → aquaporin-2). Also **constricts arteriolar smooth muscle** (V1)



Slide 4: posterior pituitary neurons

Control of ADH secretion (slide 5):

1. ⭐ **Osmolar stimulation** of hypothalamic osmoreceptors: **↑ plasma osmolality → ↑ ADH** (and vice versa).
2. **↓ Plasma volume** → fewer inhibitory impulses from **left atrial** stretch receptors → ↑ ADH.
3. **Hypotension** → carotid sinus and aortic arch **baroreceptors** → ↑↑ ADH.
4. **Higher centres**: psychological factors, stress (*[added]* also pain, nausea).
5. **Drugs:**

↑ ADH secretion	↓ ADH secretion
Acetylcholine, β-agonists, ⭐ **chlorpropamide, carbamazepine, cyclophosphamide**, **nicotine, morphine**	⭐ **Phenytoin** (diphenylhydantoin), **naloxone** (narcotic antagonists) (*[added]* alcohol)

Mnemonic for ADH-boosting drugs: "the 3 C's + N/M" *[added]* = **C**hlorpropamide, **C**arbamazepine, **C**yclophosphamide, **N**icotine, **M**orphine. They cause SIADH, and the first two are reused to treat partial central DI.

🔁 **Recap: Physiology.** ADH (supraoptic) concentrates urine in the collecting duct; oxytocin (paraventricular) ejects milk. ADH rises with ↑ osmolality, ↓ volume and ↓ BP, stress, and drugs (chlorpropamide, carbamazepine, cyclophosphamide, nicotine, morphine). It falls with phenytoin and naloxone (and alcohol).

🔗 **Link:** Remove ADH, or the kidney's response to it, and the patient cannot concentrate urine.

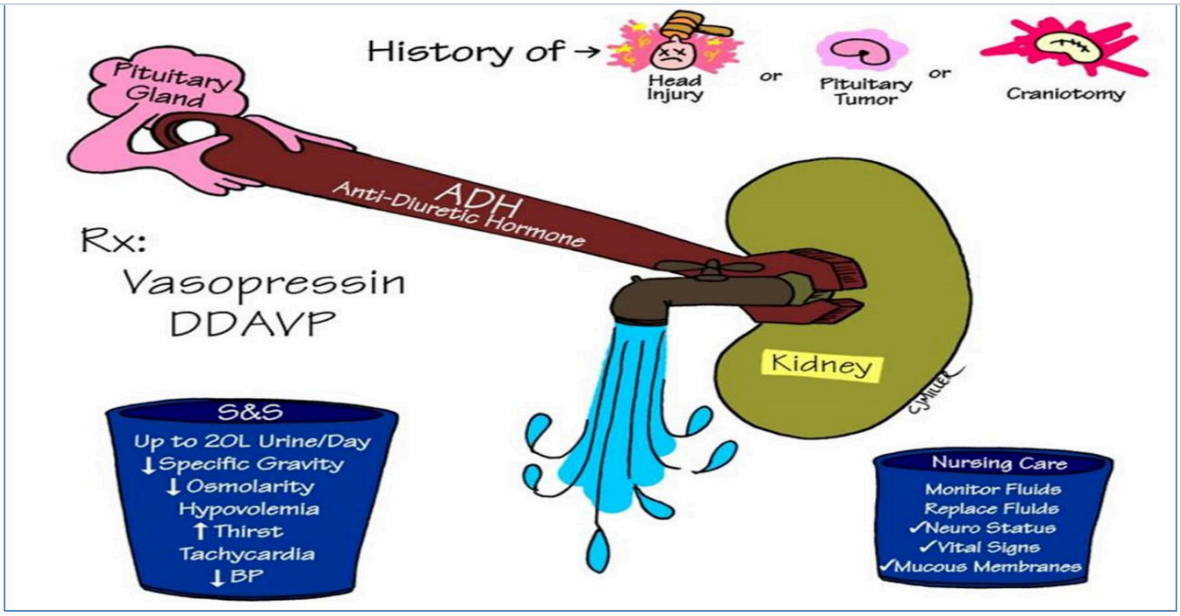
▶️ **Up next: Diabetes insipidus.**

2.2 Diabetes Insipidus (Slides 6–9) ⭐

Definition: impaired renal conservation of water from either:

- **low or absent ADH** in response to normal stimuli (**central DI**), or
- **renal unresponsiveness to ADH** (**nephrogenic DI**).

"Insipidus" means **tasteless** (dilute) urine, unlike the sweet urine of diabetes mellitus.



Slide 7: DI (head injury, pituitary tumour, craniotomy; S&S up to 20 L/day)

Causes (slide 8):

Central DI	Nephrogenic DI
Familial (rare): **autosomal dominant**, or recessive ⭐ **DIDMOAD (Wolfram syndrome)** = **DI, DM, Optic Atrophy, Deafness** (nerve)	**Familial** (*[added]* X-linked V2-receptor or AQP2 mutation)
Acquired: ⭐ **same causes as hypopituitarism** (tumour, craniopharyngioma, surgery, **head trauma**, granuloma, infection, infiltration)	**Acquired: "4 D"**
	Disturbed electrolytes: ⭐ **hypokalaemia** (1ry aldosteronism), ⭐ **hypercalcaemia** (hyperparathyroidism)
	Drugs: ⭐ **lithium**, **demeclocycline**
	Diseases of the kidney: CRF, ATN, post-transplant
	Disorders (systemic): **multiple myeloma, amyloidosis, sickle cell anaemia**

Clinical picture (slide 9):

1. ⭐ **Polyuria**: **2.5–6 L/24 h, up to 20 L/day** → **polydipsia** (*[added]* craving ice-cold water); **nocturnal enuresis** in children.
2. **Dehydration** if water intake fails (unconscious, infants, no thirst) → weakness, prostration, low-grade fever, psychological disturbance, **death** (*[added]* **hypernatraemia**).

2.3 DI: Investigations (Slides 10–12) ⭐

#	Test	Finding
1	**Urine**	↑ volume, **SG 1.001–1.004**, **fails to rise > 1.010 after fluid deprivation**
2	**X-ray sella / CT / MRI**	Exclude tumour (*[added]* MRI: loss of the **posterior pituitary "bright spot"** in central DI)
3	**Vasopressin test** (to **differentiate**)	IV vasopressin or **DDAVP** for 2–3 days. **Normal and central DI → oliguria**; **nephrogenic → no response**
4	**Hypertonic saline test** (to **confirm** DI)	2.5% NaCl 2.5 mL/kg IV stimulates osmoreceptors → **oliguria if ADH system intact**
5	⭐ **Water deprivation (dehydration) test** (to **differentiate**)	See below

Water deprivation test (slide 11):

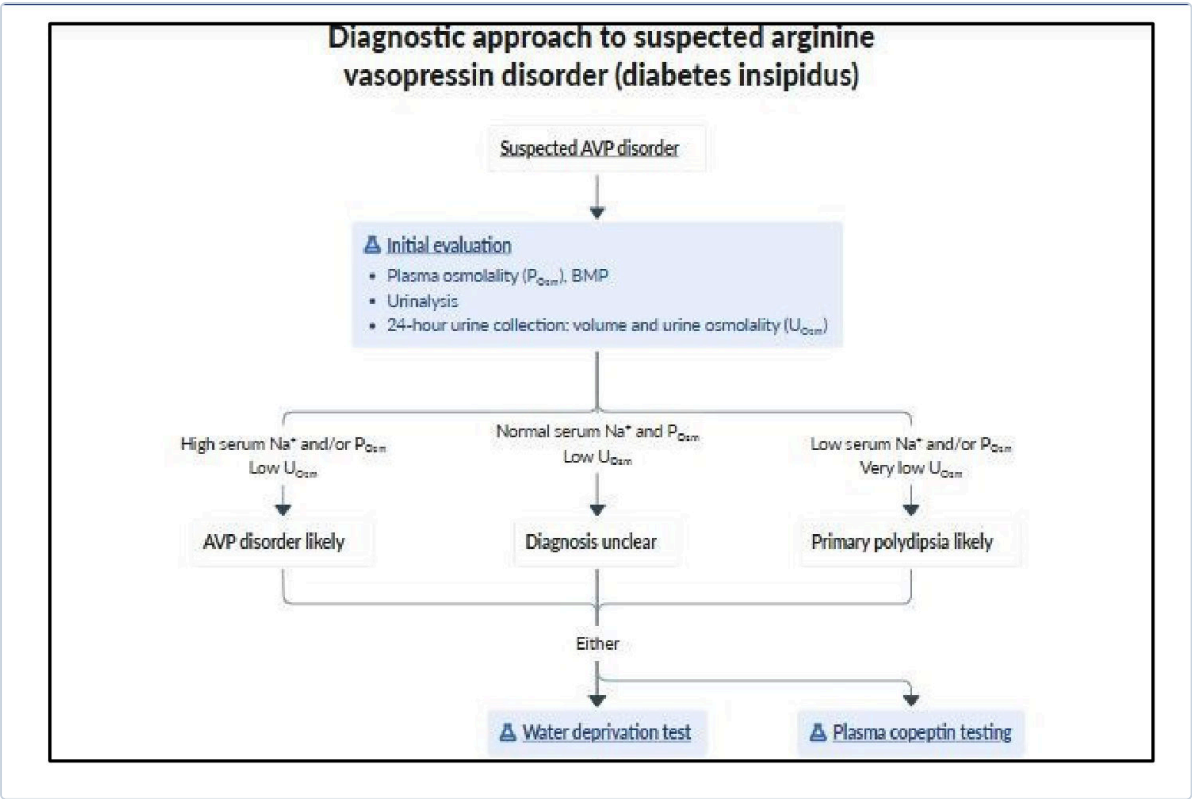
- **Deprive water for 4–18 h**, measuring urine osmolality hourly.
- **End point: successive urine samples of the same osmolality (plateau)**, or **5% body-weight loss**.
- Then give **5 U aqueous ADH SC** (or desmopressin) and measure urine osmolality **1 h later**.

Group	Urine osm at end of deprivation	After ADH
Normal / psychogenic polydipsia	**↑ (concentrates)**	**No further ↑**
⭐ **Central DI**	**Low**	⭐ **↑ (responds)**
⭐ **Nephrogenic DI**	**Low**	⭐ **Low (no response)**

[added] Typical numbers: a normal person reaches urine osm > 800. In central DI it rises > 50% after DDAVP; in nephrogenic DI it rises < 10%. Partial forms fall in between. Stop the test if plasma Na > 145–150 or weight loss > 3–5%.

Diagnostic approach to suspected AVP disorder (slide 12):

- **Initial:** plasma osmolality, BMP, urinalysis, **24-h urine volume and osmolality**.
- Then: - **High Na/P_osm + low U_osm** → AVP disorder likely. - **Normal Na + low U_osm** → unclear. - **Low Na/P_osm + very low U_osm** → **primary polydipsia** likely.
- Then **water deprivation test** or ⭐ **plasma copeptin** (*[added]* copeptin is a stable AVP surrogate; hypertonic-saline-stimulated copeptin is now preferred).



Slide 12: diagnostic approach to suspected AVP disorder

Polyuria (≥ 3 L/day): glucose first, then the water deprivation test

Polyuria + polydipsia
Check glucose (DM), Ca^{2+} (↑), K^+ (↓), creatinine, drugs (lithium)

Water deprivation 4–18 h (stop at plateau or 5% weight loss)
Then give ADH / desmopressin and recheck urine osmolality

Normal / PSYCHOGENIC polydipsia

Deprivation: urine osm ↑ (concentrates)
After ADH: no further rise

Plasma Na/osm LOW-normal
No nocturia, emotional disturbance

Rx: restrict fluid, psychiatric care

CENTRAL DI (no ADH)

Deprivation: urine stays DILUTE
After ADH: urine osm RISES ↑

Head trauma, surgery, tumour, hypopituitarism causes, Wolfram

Rx: DESMOPRESSIN (DDAVP)
± chlorpropamide, carbamazepine, clofibrate

NEPHROGENIC DI (no response)

Deprivation: urine stays DILUTE
After ADH: NO change

"4 D": Disturbed electrolytes (↓K, ↑Ca), Drugs (LITHIUM, demeclocycline), Diseased kidney, Disorders (MM, SCA)

Rx: THIAZIDE; AMILORIDE (lithium)

Opposite problem, SIADH: too much ADH → euvolaemic hyponatraemia (Na < 135), plasma osm < 275, urine osm > 100, urine Na > 20–30 → restrict water (< 500 mL below output), demeclocycline, 3% saline if severe (≤ 10–12 mmol/L/24 h)

Added diagram: polyuria and the water deprivation test

2.4 DI: Treatment (Slide 13) ⭐

Central DI	Nephrogenic DI
Hormone replacement: aqueous vasopressin (Pitressin) SC; ⭐ **desmopressin (DDAVP)** oral (Minirin), sublingual or intranasal. **No pallor, angina, colic, bronchospasm or uterine contraction** (selective V2 action). **Vasopressin tannate in oil** IM	⭐ **Thiazides** (HCTZ, chlorthalidone): mild volume contraction → ↑ proximal Na/water reabsorption → less delivered to the loop → ↓ water excretion (paradoxical antidiuresis)
Non-hormonal (↑ ADH release/effect) for **partial** central DI: **chlorpropamide, clofibrate, carbamazepine**	⭐ **Amiloride: drug of choice in lithium-induced DI** (blocks lithium entry via ENaC)

Other measures:

- Treat the **cause**.
- Diet: **liberal fluids**; **avoid a high-purine diet** (exaggerates polyuria; *[added]* a **low-solute** diet — low salt, low protein — is the modern advice).

[added] Also stop lithium if possible; correct K^+/Ca^{2+}; NSAIDs (indomethacin) help in nephrogenic DI. Watch for hyponatraemia with desmopressin overdose.

2.5 DDx of Polyuria (≥ 3 L/day) (Slides 14–15)

Group	Causes
Renal	**CRF** (chronic pyelonephritis, chronic GN) · **diuretic phase of ATN** · **excess diuretics** · **nephrogenic DI** · **RTA** and **De Toni–Fanconi**
Endocrine	⭐ **Diabetes mellitus** (**high SG**, glycosuria, high glucose) · **DI** · **adrenal**: Cushing's, Conn's, Addison's · ⭐ **hyperparathyroidism ("calcium diabetes")** and other hypercalcaemia · **thyrotoxicosis**
Functional	**Excess fluid** (coffee, tea, cola, beer) · ⭐ **psychogenic (hysterical) polydipsia**: polyuria + polydipsia, **NO nocturia**, emotional disturbance (e.g. anorexia nervosa), **low urine osm with a NORMAL response to dehydration** · **winter**
Miscellaneous	**Multiple myeloma**, **vit D intoxication** · **transient** polyuria after migraine, epilepsy, paroxysmal tachycardia, asthma, intermittent hydronephrosis

Key split: water diuresis (urine osm < 300: DI, psychogenic) vs **solute diuresis** (osm > 300: glucose, urea, mannitol, saline) *[added]*.

🔁 **Recap: DI.**

- **Types:** central (no ADH: hypopituitary causes, head trauma, surgery, Wolfram) vs nephrogenic (no response: ↓K, ↑Ca, lithium, demeclocycline, CRF/ATN, myeloma, amyloid, sickle cell).
- **Features:** 2.5–20 L/day of dilute urine (SG < 1.005), thirst, dehydration/hypernatraemia if fluids are restricted.
- **Water deprivation then ADH:** normal/psychogenic concentrate; central rises only after ADH; nephrogenic never rises. Copeptin is the modern test.
- **Treatment:** central → desmopressin (or chlorpropamide/carbamazepine/clofibrate if partial); nephrogenic → thiazide, amiloride for lithium.
- **Polyuria DDx:** DM (high SG), hypercalcaemia, hypokalaemia, renal, psychogenic (no nocturia).

🔗 **Link:** The mirror image: ADH secreted when it shouldn't be, so water is retained.

▶️ **Up next: SIADH.**

2.6 SIADH (Slides 16–20) ⭐

Mechanism: ↑ ADH secretion **not in response to normal physiological stimuli** → **↓ ability to excrete water** → **dilutional (euvolaemic) hyponatraemia**.

Causes (slide 17):

Group	Examples
⭐ **Malignancy**	**Oat (small) cell carcinoma of the lung** (ectopic ADH)
Non-malignant lung disease	**Pneumonia, TB**
CNS	**Cerebrovascular accidents** (*[added]* trauma, meningitis, SAH)

Group	Examples
Drugs	⭐ **Chlorpropamide, cyclophosphamide, vincristine** (*[added]* **SSRIs, carbamazepine**, TCAs, MDMA)
Miscellaneous	**Hypothyroidism**, acute psychosis (*[added]* pain, nausea, post-op)

Clinical picture (water intoxication; slide 18):

- **Chronic** hyponatraemia may be **asymptomatic**; CNS symptoms are less common.
- **Mild:** anorexia, nausea, vomiting, headache, cramps.
- **Moderate:** weakness, lethargy, confusion.
- ⭐ **Severe: seizures, altered consciousness**.
- ⭐ **Normotension, NO oedema** (euvolaemic) + features of the cause.
- **Pearl:** SIADH patients are **euvolaemic, normotensive and have no oedema**. A hyponatraemic patient **with oedema** suggests another condition (e.g. **CHF**).

Laboratory (slide 19):

- **Serum Na < 130 mEq/L** (dilutional, euvolaemic)
- **↓ Plasma osmolality < 270 mOsm/kg**
- **↑ Urine osmolality > 100 mOsm/kg** (inappropriately concentrated)

Diagnostic criteria (slide 19 table):

Criterion	Value
Hyponatraemia	Serum Na **< 135 mEq/L**
Hypo-osmolality	Serum osm **< 275 mOsm/kg**
Euvolaemia	No signs of hypo- or hypervolaemia
Concentrated urine	Urine osm **> 100 mOsm/kg**
Elevated urinary Na	Urine Na **> 20–30 mEq/L**

[added] Also required: normal thyroid, adrenal and renal function, and no diuretics. Typical extras: low urea and uric acid.

Treatment (slide 20):

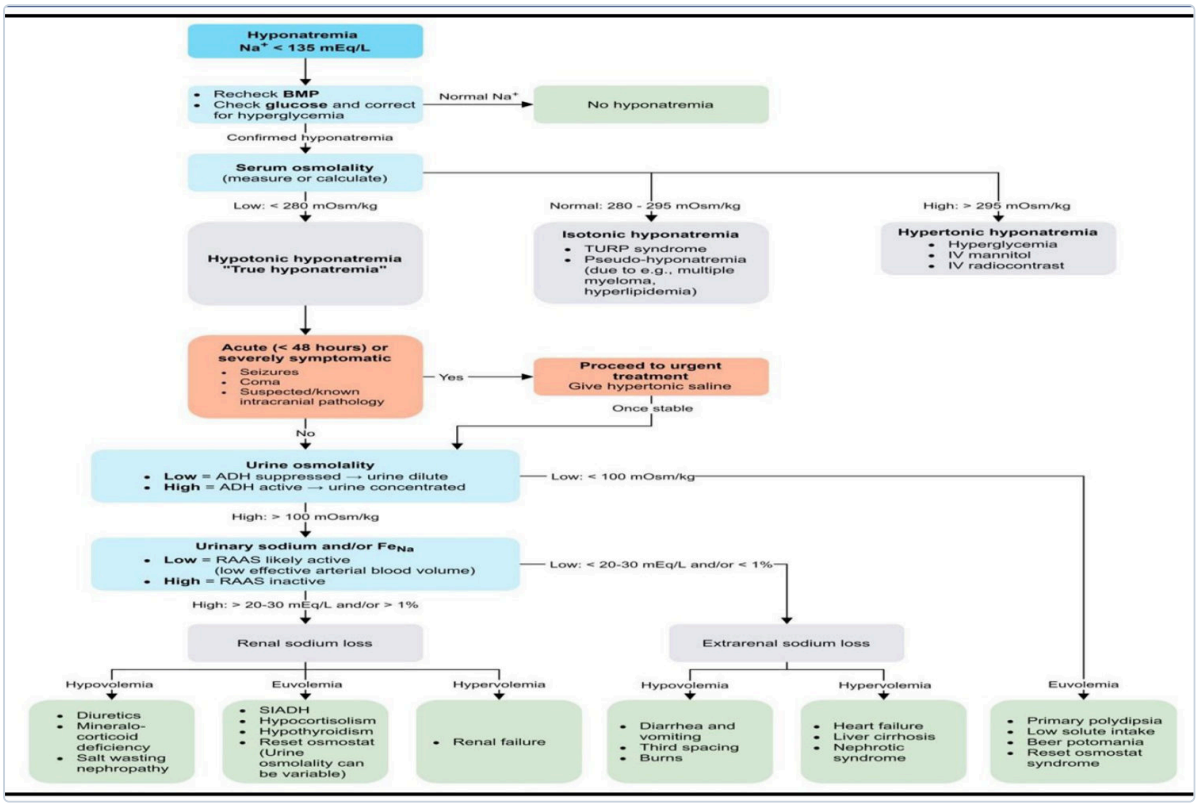
- **Treat the cause.**
- ⭐ **Restrict water.** Pearl: daily intake should be **500 mL less than daily urine output**.
- **Drugs:** - **demeclocycline** (induces nephrogenic DI) - **phenytoin** - **naloxone** - *[added]* **tolvaptan** (V2 antagonist), **urea**, oral salt + loop diuretic
- ⭐ **Hypertonic saline (5%; *[added]* usually 3%) in emergencies** (seizures, coma).
- ⭐ **Correction limit:** in chronic hyponatraemia the maximum (not the goal) is **10–12 mEq/L per 24 h**; **8 mEq/L per 24 h** if at high risk of **osmotic demyelination**.
- ⭐ **Caution:** over-correction of chronic hyponatraemia → **osmotic demyelination syndrome** (central pontine myelinolysis).

> **Mnemonic: "From low to high, the pons will die; from high to low, the brain will blow."** *[added]* Correcting hyponatraemia too fast causes ODS; correcting hypernatraemia too fast causes cerebral oedema.

2.7 Diagnostic Approach to Hyponatraemia (Slide 21) ⭐

1. **Na < 135** → repeat; check **glucose** and correct for **hyperglycaemia**.

2. **Serum osmolality:** - **High > 295 → hypertonic hyponatraemia**: **hyperglycaemia**, IV mannitol, IV radiocontrast. - **Normal 280–295 → isotonic**: **TURP syndrome**, **pseudohyponatraemia** (**multiple myeloma, hyperlipidaemia**). - **Low < 280 → hypotonic ("true") hyponatraemia.**
3. **Acute (< 48 h) or severely symptomatic?** (seizure, coma, known intracranial pathology) → **urgent hypertonic saline**, then continue the work-up once stable.
4. **Urine osmolality:** - **< 100 → ADH suppressed (dilute urine) → euvolaemic**: **primary polydipsia**, **low solute intake**, **beer potomania**, reset osmostat. - **> 100 → ADH active** → check **urine Na / FeNa**.
5. **Urine Na:** - **Low < 20–30 mEq/L (FeNa < 1%) → extrarenal Na loss**: - **hypovolaemic**: diarrhoea, vomiting, third spacing, burns - **hypervolaemic**: **heart failure, cirrhosis, nephrotic** - **High > 20–30 (FeNa > 1%) → renal Na loss**: - **hypovolaemic**: *[read from figure]* diuretics, mineralocorticoid deficiency, salt-wasting nephropathy - **euvolaemic**: ⭐ **SIADH**, **hypocortisolism**, **hypothyroidism**, reset osmostat - **hypervolaemic**: **renal failure**



Slide 21: diagnostic approach to hyponatraemia

> 🔁 **Recap: SIADH and hyponatraemia.**
>
> - **SIADH:** ADH without a stimulus. Causes: oat cell cancer, pneumonia/TB, CNS events, drugs (chlorpropamide, cyclophosphamide, vincristine, SSRIs, carbamazepine), hypothyroidism. Euvolaemic, normotensive, no oedema. Na < 135, Posm < 275, Uosm > 100, UNa > 20–30.
> - **Treatment:** restrict fluid (500 mL below output); demeclocycline/tolvaptan; hypertonic saline if severe, correcting ≤ 10–12 mmol/L/24 h (ODS risk).
> - **Algorithm:** serum osm (hyper → glucose; iso → pseudo/TURP; hypo →) → acute or severe? → urine osm (< 100 = polydipsia/potomania) → urine Na and volume.

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3. High-Yield Comparison Tables

Table A: DI vs SIADH

	DI	SIADH
ADH effect	**↓** (central) or ignored (nephrogenic)	**↑ inappropriately**
Urine volume	**↑↑** (2.5–20 L)	↓ / normal
Urine osm / SG	**Low** (SG < 1.005)	**High** (> 100; inappropriately concentrated)
Serum Na	**N or ↑** (hypernatraemia if no water)	**↓** (< 135)
Plasma osm	N / ↑	**↓** (< 275)
Volume	N / dehydrated	**Euvolaemic**, no oedema
Treatment	Desmopressin (central); thiazide/amiloride (nephrogenic)	**Fluid restriction**, demeclocycline/tolvaptan, hypertonic saline

Table B: Central vs Nephrogenic DI vs Psychogenic Polydipsia

	Central	Nephrogenic	Psychogenic
Plasma Na/osm	High-normal	High-normal	**Low-normal**
Deprivation urine osm	**Low**	**Low**	**↑**
After ADH	**↑ (> 50%)**	**No change**	No further ↑
Nocturia	Yes	Yes	**No**
Causes	Hypopituitary causes, trauma, surgery, Wolfram	**↓K, ↑Ca, lithium, demeclocycline**, CRF, myeloma, amyloid, SCA	Psychiatric
Treatment	**Desmopressin**	**Thiazide**; **amiloride (lithium)**	Fluid restriction

Table C: Drugs on Both Sides

Drug	Effect	Use / harm
Chlorpropamide, carbamazepine	↑ ADH release/action	Treat **partial central DI**; **cause SIADH**
Cyclophosphamide, vincristine	↑ ADH	Cause SIADH
Lithium, demeclocycline	Block ADH action	**Cause nephrogenic DI**; demeclocycline **treats SIADH**
Phenytoin, naloxone	↓ ADH release	Listed for SIADH treatment
Thiazides	Volume contraction	**Treat nephrogenic DI**; *[added]* cause hyponatraemia

Table D: Hyponatraemia by Volume Status *(slide 21)*

Volume	Urine Na low (< 20–30)	Urine Na high (> 20–30)
Hypovolaemic	GI loss, third spacing, burns	**Diuretics**, mineralocorticoid deficiency, salt wasting

Volume	Urine Na low (< 20–30)	Urine Na high (> 20–30)
Euvolaemic	(U osm < 100: **polydipsia, beer potomania, low solute**)	**SIADH**, **hypothyroidism**, **hypocortisolism**
Hypervolaemic	**HF, cirrhosis, nephrotic**	**Renal failure**

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
ADH ↑ with osmolality	Also ↑ with **volume loss / hypotension** (overrides osmolality), stress, nausea, drugs
Polyuria + polydipsia = DI	**DM first** (high SG + glucose); then hypercalcaemia, hypokalaemia, psychogenic
Water deprivation diagnoses DI	Normal **and psychogenic** both concentrate. Only DI stays dilute
Desmopressin treats DI	**Only central**. Nephrogenic → **thiazide** (paradoxical) or **amiloride** for lithium
Thiazides are diuretics	In nephrogenic DI they **reduce** urine output
Hyponatraemia → give saline	**SIADH → restrict water**. Normal saline can **worsen** it (Na excreted, water kept) *[added]*
Correct Na quickly	**Chronic → ≤ 10–12 (≤ 8 high risk) mmol/L/24 h**, or ODS
SIADH = euvolaemic	**Oedema → think HF, cirrhosis, nephrotic**; **dehydration → think hypovolaemia**
Low Na = low osmolality	**Hyperglycaemia** (hypertonic) and **myeloma/lipids** (pseudo, isotonic)
Urine osm in polydipsia	**< 100** (maximally dilute). SIADH **> 100**
Demeclocycline is an antibiotic	Used here **to cause nephrogenic DI** deliberately (SIADH treatment)

5. Clinical Correlations

- **Polyuria after pituitary surgery or head trauma** → **central DI** (*[added]* can be triphasic: DI → SIADH → DI).
- **Bipolar patient on lithium with polyuria** → **nephrogenic DI** → **amiloride**; consider stopping lithium.
- **Polyuria + bone aches + Ca 13 + high PTH** → **primary HPT → hypercalcaemia → nephrogenic DI** (lecture case).
- **Heavy smoker with Na 118, euvolaemic, concentrated urine** → **SIADH from small cell lung cancer**.
- **Elderly woman started on an SSRI, now confused, Na 121** → drug-induced **SIADH** → stop the drug, restrict fluid.
- **Beer drinker with little food intake, Na 115, urine osm 60** → **beer potomania** (low solute).
- **Patient with schizophrenia drinking 10 L/day, urine osm 80, concentrates on deprivation** → **psychogenic polydipsia**.
- **Hyponatraemia with oedema and ascites** → cirrhosis/HF (hypervolaemic), not SIADH.
- **Rapid correction of Na from 110 to 130 in a day**, then quadriparesis, dysarthria and "locked-in" state 3 days later → **osmotic demyelination**.
- **Na 128 with glucose 900** → **pseudo/hypertonic hyponatraemia**: correct for glucose (*[added]* +1.6–2.4 mmol/L per 100 mg/dL above 100).

6. Rapid Revision (Last-Day)

1. **ADH = supraoptic; oxytocin = paraventricular** → stored in the posterior pituitary.
2. ADH ↑ water reabsorption in the collecting duct (and vasoconstriction). Oxytocin = milk ejection.
3. ADH stimuli: **↑ osmolality**, **↓ volume** (LA receptors), **hypotension** (baroreceptors), stress, drugs (**chlorpropamide, carbamazepine, cyclophosphamide, nicotine, morphine**, ACh, β-agonists). Inhibited by **phenytoin, naloxone**.
4. **DI** = impaired renal water conservation: **central** (↓ ADH) or **nephrogenic** (renal resistance).
5. Central: familial (AD; **Wolfram DIDMOAD**) or acquired (**hypopituitarism causes**).
6. Nephrogenic "4 D": **↓K, ↑Ca**; **lithium, demeclocycline**; **CRF, ATN, transplant**; **myeloma, amyloid, SCA**.
7. Polyuria **2.5–6 L (up to 20 L)**, polydipsia, enuresis; dehydration can kill.
8. Urine **SG 1.001–1.004**, won't exceed **1.010** after deprivation.
9. **Water deprivation 4–18 h** to plateau or 5% weight loss → **5 U ADH**: normal/psychogenic concentrate; **central ↑ only after ADH**; **nephrogenic never**.
10. Hypertonic saline test **confirms** DI; vasopressin test **differentiates**; copeptin is the modern alternative.
11. Treatment: central → **DDAVP** (oral, sublingual or intranasal; no pallor, angina, colic, bronchospasm or uterine contraction), Pitressin; partial → **chlorpropamide, clofibrate, carbamazepine**. Nephrogenic → **thiazide**; **amiloride for lithium**. Liberal fluids; avoid purines.
12. Polyuria DDx: renal (CRF, ATN recovery, diuretics, RTA, Fanconi), endocrine (**DM**, DI, Cushing's, Conn's, Addison's, **hyperPTH**, thyrotoxicosis), functional (**psychogenic: no nocturia, normal response to deprivation**), myeloma, vit D.
13. **SIADH causes: oat cell carcinoma**, pneumonia, TB, CVA, **chlorpropamide, cyclophosphamide, vincristine**, hypothyroidism, psychosis.
14. SIADH = **euvolaemic, normotensive, no oedema**. Mild: GI symptoms, headache, cramps. Moderate: lethargy, confusion. Severe: **seizures, coma**.
15. Labs: **Na < 135 (< 130 on slide 19), Posm < 275 (< 270), Uosm > 100, UNa > 20–30**.
16. Treatment: cause; **fluid restriction (500 mL below urine output)**; demeclocycline, phenytoin, naloxone; **hypertonic saline in emergency**; max **10–12 mmol/L/24 h (8 if high risk)** → avoid **ODS**.
17. Algorithm: **Sosm** high → **hyperglycaemia/mannitol**; normal → **pseudo (myeloma, lipids)/TURP**; low → acute/severe? → **Uosm < 100** (polydipsia, potomania) → **UNa** + volume.

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7. Exam-Style Questions

Q1 (lecture case, slides 22–24). A 63-year-old woman (T2DM, depression, HTN; on lisinopril, escitalopram, metformin) has 4 weeks of urinary frequency, nocturia, **bone aches**, nausea, fatigue, and thirst (5–6 L/day). Dry mucosa. Labs: Na 151, **Ca 13 mg/dL**, glucose 101, **PTH 130 pg/mL**, serum osm 306, **urine osm 255**. After water deprivation: Posm 309, Uosm 256. **One hour after desmopressin: Uosm 259 (no change).** Cause of the urinary frequency? a) **Nephrogenic DI** · b) HHS · c) Beer potomania · d) Psychogenic polydipsia · e) Central DI · f) SIADH

Answer: a. Nephrogenic diabetes insipidus, secondary to **hypercalcaemia from primary hyperparathyroidism** (high Ca + high PTH, bone aches). *[reasoning added; key not on slides]*

- Urine stays dilute after deprivation and **does not respond to desmopressin**.
- **HHS** is excluded: glucose is 101.
- **Psychogenic polydipsia** gives low Na and urine that concentrates.
- **Central DI** would respond to desmopressin.
- **Beer potomania** and **SIADH** cause **hypo**natraemia.
- Escitalopram (an SSRI) is a SIADH distractor.

Q2. A man on long-term lithium has 8 L/day of dilute urine. Water deprivation: urine stays dilute; no rise after DDAVP. What is the best drug? A. Desmopressin · B. Chlorpropamide · C. **Amiloride** · D. Demeclocycline · E. Carbamazepine

Answer: C.

- **Amiloride is the drug of choice in lithium-induced nephrogenic DI** (slide 13).
- Desmopressin, chlorpropamide and carbamazepine act on the ADH axis, which isn't the problem.
- Demeclocycline *causes* nephrogenic DI.

Q3. A smoker with a central lung mass: Na 121, serum osm 255, urine osm 480, urine Na 45, no oedema, BP normal, normal TSH and cortisol. Diagnosis and first-line management? A. Hypovolaemic hyponatraemia; normal saline · B. **SIADH; fluid restriction** · C. Psychogenic polydipsia; psychiatric review · D. Cirrhosis; diuretics · E. Adrenal insufficiency; hydrocortisone

Answer: B.

- **SIADH** (oat cell carcinoma): **euvolaemic**, Posm < 275, Uosm > 100, UNa > 20–30 (slide 19).
- Restrict fluid to **500 mL below urine output**.

Q4. A patient with Na 112 had a seizure. Hypertonic saline is started. What is the maximum safe correction over the first 24 h in chronic hyponatraemia? A. 25 mmol/L · B. 20 mmol/L · C. **10–12 mmol/L (8 if high risk)** · D. 2 mmol/L · E. No limit after a seizure

Answer: C (slide 20 pearl).

- Faster correction risks **osmotic demyelination**.

Q5. Which finding distinguishes **psychogenic polydipsia** from DI on the water deprivation test? A. Low urine osmolality at baseline · B. **Urine concentrates with deprivation; no further rise after ADH** · C. Urine rises only after ADH · D. No response at all · E. High plasma Na

Answer: B (slide 11).

- C = central DI; D = nephrogenic DI.

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8. Exam Pitfalls & Traps

- **Nucleus mix-up:** this lecture (slide 4) is right: **ADH = supraoptic, oxytocin = paraventricular**. The Pituitary I histology slide reverses them.
- **SIADH cut-offs differ on the same slide** (slide 19): text says **Na < 130, Posm < 270**; the criteria table says **Na < 135, Posm < 275**. Use the **table (135/275)** for diagnostic criteria.
- **Hypertonic saline:** the slide says **5%**. *[added]* Most current protocols use **3%** (100–150 mL boluses). Quote the slide if asked about the lecture.
- **Water deprivation interpretation:** psychogenic polydipsia **concentrates** (like normal people). Don't call it DI because baseline urine is dilute.
- **Thiazide in DI** reduces urine output. It is counter-intuitive, which is why it is tested.
- **Lithium DI → amiloride** (not just "thiazide").
- **Chlorpropamide and carbamazepine** appear on both lists: they **treat partial central DI** and **cause SIADH**. Same mechanism (↑ ADH).
- **Demeclocycline** treats SIADH by **inducing nephrogenic DI**.
- **"4 D" for nephrogenic DI** includes **hypokalaemia AND hypercalcaemia**. Classic MCQ links: Conn's → polyuria; hyperparathyroidism → polyuria.
- **Normal saline in SIADH** can **lower** Na further (urine osm > saline osm) *[added]*.
- **Pseudohyponatraemia** (myeloma, hyperlipidaemia) and **hyperglycaemia** must be excluded first by measuring serum osmolality.
- **Slide 10 numbering is out of order** (1, 2, 4, 3). Content is unaffected.
- **"Avoid purine-rich diet"** (slide 13) is the lecturer's point. *[added]* Modern advice is a **low-solute** (low salt, low protein) diet.